

THE WEST AFRICAN EXAMINATIONS COUNCIL

BASIC EDUCATION CERTIFICATE EXAMINATION -- JUNE 2008

MATHEMATICS -- PAPER 2

Duration: 1 Hour 30 Minutes | Level: JHS 3 / BECE | Code: MATH 01/02 - P2

INSTRUCTIONS TO CANDIDATES

Answer all four questions in Section A and any three questions in Section B. All working must be shown clearly in your answer booklet. Credit will be given for clarity of expression and orderly presentation.

PART II: ESSAY & STRUCTURED QUESTIONS WITH WAEC RUBRICS

QUESTION 1: Algebra and Number Operations

A school bookshop purchased a number of exercise books and pens for distribution to students.

(a) Simplify: $2\frac{1}{3} - 1\frac{3}{4} + 5\frac{5}{6}$. [4 marks]

Model Answer / WAEC Rubric: Convert to improper fractions: $\frac{7}{3} - \frac{7}{4} + \frac{5}{6}$. Common denominator = 12. $(\frac{28}{12} - \frac{21}{12} + \frac{10}{12}) / 12 = \frac{17}{12} = 1\frac{5}{12}$. [M1, A2]

(b) Solve the simultaneous equations: $2x + 3y = 13$ and $3x - y = 3$. [6 marks]

Model Answer / WAEC Rubric: From eqn (2), $y = 3x - 3$. Substitute into (1): $2x + 3(3x - 3) = 13 \Rightarrow 11x = 22 \Rightarrow x = 2$. Then $y = 3(2) - 3 = 3$. Therefore $x = 2$, $y = 3$. [M2, A2, B2]

(c) Factorize completely: $6ax - 9ay + 4bx - 6by$. [4 marks]

Model Answer / WAEC Rubric: Group terms: $3a(2x - 3y) + 2b(2x - 3y) = (3a + 2b)(2x - 3y)$. [M2, A2]

QUESTION 2: Geometry and Mensuration

A cylindrical water reservoir in a village has an internal radius of 1.4 m and a depth of 5 m.

(a) Calculate the volume of the reservoir in cubic metres. [Take $\pi = \frac{22}{7}$] [4 marks]

Model Answer / WAEC Rubric: Volume = $\pi \times r^2 \times h = (\frac{22}{7}) \times (1.4)^2 \times 5 = \frac{22}{7} \times 1.96 \times 5 = 30.8 \text{ m}^3$. [M2, A2]

(b) How many litres of water can the reservoir hold when completely full? ($1 \text{ m}^3 = 1,000$ litres) [3 marks]

Model Answer / WAEC Rubric: Capacity = $30.8 \times 1,000 = 30,800$ litres. [M1, A2]

(c) If water is pumped out at a rate of 140 litres per minute, how long in hours will it take to empty the reservoir? [5 marks]

Model Answer / WAEC Rubric: Time = $30,800 / 140 = 220$ minutes = 3 hours 40 minutes (or $3\frac{2}{3}$ hours). [M2, A3]

QUESTION 3: Statistics and Data Analysis

The marks obtained by 30 candidates in a test were recorded as follows: 12, 14, 15, 12, 16, 18, 14, 15, 17, 12, 16, 15, 14, 15, 18, 19, 15, 14, 16, 15, 12, 17, 18, 15, 14, 16, 15, 17, 14, 15.

(a) Construct a frequency distribution table for the data. [4 marks]

Model Answer / WAEC Rubric: Table showing Marks (12, 14, 15, 16, 17, 18, 19), Tallies, and Frequencies (4, 6, 9, 4, 3, 3, 1; total = 30). [M2, A2]

(b) State the mode of the distribution. [2 marks]

Model Answer / WAEC Rubric: The mode is 15 (highest frequency = 9). [B2]

(c) Calculate the mean mark of the candidates. [6 marks]

Model Answer / WAEC Rubric: Sum $\Sigma fx = (12 \times 4) + (14 \times 6) + (15 \times 9) + (16 \times 4) + (17 \times 3) + (18 \times 3) + (19 \times 1) = 48 + 84 + 135 + 64 + 51 + 54 + 19 = 455$. Mean = $455 / 30 = 15.17$ marks. [M3, A3]

QUESTION 4: Plane Geometry and Vectors

In triangle ABC, $AB = 8 \text{ cm}$, angle $ABC = 90^\circ$ and angle $BAC = 30^\circ$.

(a) Using ruler and compasses only, construct triangle ABC. [5 marks]

Model Answer / WAEC Rubric: Accurate construction of line $AB = 8 \text{ cm}$, 90° perpendicular at B, 30° angle at A, and intersection C. [M3, A2]

(b) Measure the length of the hypotenuse AC. [2 marks]

Model Answer / WAEC Rubric: $AC = 9.24 \text{ cm (+/- } 0.1 \text{ cm)}$. [A2]

(c) Given vector $p = (4, -3)$ and vector $q = (-2, 5)$, find the magnitude of $(p + q)$. [5 marks]

Model Answer / WAEC Rubric: $p + q = (4 - 2, -3 + 5) = (2, 2)$. Magnitude = $\sqrt{2^2 + 2^2} = \sqrt{8} = 2\sqrt{2} = 2.83 \text{ units}$. [M3, A2]